

Sparta Timesheet Approver

User Manual

Rail Delivery Group - QA Team

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About This Document

This manual describes the automated Sparta Timesheet Approver process used by the Rail Delivery Group QA team. It is written for any reader who needs to understand, operate, or audit the timecard approval workflow - no technical background is required.

The tool compares employee timesheets submitted in Sparta Global against work logs recorded in JIRA, applies a defined set of business rules to each record, and automatically approves or rejects timecards on the manager's behalf. A PDF audit report is generated for every timecard processed.

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1. What Is This?

Every fortnight, employees submit their timesheets through the Sparta Global portal. As the approving manager, you need to verify that the hours each employee has claimed match what they actually logged in JIRA - the project tracking tool where employees record their daily work.

This tool automates that entire process. Instead of manually logging into both systems and comparing numbers, the agent does it for you:

- Asks you for the resource name (employee) to process - this is always the first step
- Logs into Sparta and retrieves each pending timecard for that resource
- Pulls the employee's JIRA work logs directly using the JIRA API
- Applies a set of rules to decide whether to approve or reject
- Actions the decision inside Sparta on your behalf
- Produces a PDF audit report for every timecard processed

2. Prerequisites

The following must be in place before the tool can run:

Prerequisite 1 - Resource Name - Required to Start

Before the tool begins processing any timecards, it will always ask you for the name of the resource (employee) whose timecards you want to review. This is mandatory.

You must provide the employee's full name exactly as it appears in Sparta, for example:

Francis Adewale

Olayemi Ojo

Raihan Kamal

The tool will not proceed until a valid resource name is provided. This ensures timecards are always processed intentionally and for the correct individual.

Prerequisite 2 - Sparta Account

You must have a Sparta Global manager account with permission to approve timecards.

- Login URL: <https://customerspartaglobal.my.site.com/Client/s/login/>
- Username format: `yourname@company.com.sparta`

Prerequisite 3 - JIRA Access

You need a JIRA account at raildeliverygroup.atlassian.net with read access to employee worklogs.

- JIRA email: your normal work email (e.g. carl.solomon@raildeliverygroup.com)
- JIRA API token: a personal access token generated from your Atlassian account.

To create a JIRA API token:

1. Go to: <https://id.atlassian.com/manage-profile/security/api-tokens>
2. Click Create API token
3. Give it a name (e.g. Sparta Approver) and copy the token

Store it securely - you only see it once.

Prerequisite 4 - Employee JIRA Account IDs

Each employee has a unique internal JIRA identifier called an Account ID. This is different from their name or email address. The tool needs this to pull their work logs.

For known employees, these are pre-stored in the configuration. When processing a new employee for the first time, the agent will look up their Account ID automatically using their name.

Prerequisite 5 - Python Environment (Local Execution Only)

Only required if running the pipeline scripts on your own machine rather than through the agent.

- Python 3.9 or later
- `pip install fpdf2 playwright`
- `playwright install chromium`

3. How It Works - The 10-Step Process

Every timecard goes through exactly 10 steps, in order, without exception. No steps are batched or skipped.

Step 0 - Provide the Resource Name

Before any timecard is opened, the agent prompts you to enter the resource name (employee) you want to process. No timecards will be retrieved until this is provided. Once confirmed, the agent proceeds with all pending timecards for that employee.

Step 1 - Open the Timecard in Sparta

The agent navigates to the Sparta portal, selects the Timecards menu, and opens the individual timecard record for the employee being reviewed.

Step 2 - Screenshot the Timecard

A screenshot of the open timecard is taken and saved. This captures the employee name, start and end date of the period, total hours submitted, and the project name. This screenshot is included in the audit report.

Step 3 - Extract the Key Details

The following key details are read from the timecard:

Field	Example
Employee name	Olayemi Ojo
Period start	11/05/2026
Period end	17/05/2026
Hours submitted to Sparta	40.00
Project	20250814 - Rail Settlement Plan LTD

Step 4 - Find the Employee in JIRA

Using the employee's name, the agent looks up their unique JIRA Account ID. This identifier is required to fetch their work logs in the next step.

Step 5 - Pull JIRA Work Logs via API

The agent calls the JIRA REST API directly (not the JIRA website) to retrieve every work log entry the employee has made during the timecard period. Each log entry includes the date, JIRA issue key, task name, and hours logged.

Step 6 - Apply the Decision Rules

The decision rules are applied (see Section 4 below). The agent compares Sparta hours against JIRA total hours, accounting for any leave or holiday entries.

Step 7 - Action the Decision in Sparta

Based on the decision, the agent clicks Approve or Reject on the Sparta timecard page and confirms the action in the dialog. No manual intervention is needed.

Step 8 - Screenshot the Confirmation

Once the action is confirmed, a screenshot of the updated timecard is taken, capturing the new status (Approved or

Rejected) as proof the action was completed.

Step 9 - Generate the Audit Report

A PDF audit report is generated automatically (see Section 5). It is uploaded and a download link is provided. The record is also saved to the audit log.

Step 10 - Report Back and Move On

The agent reports the outcome - timecard ID, employee, period, decision, and a link to the PDF. It then moves on to the next pending timecard and repeats all 10 steps from the beginning.

4. The Decision Rules

The core logic compares three figures:

Sparta Hours - what the employee submitted in the Sparta portal

JIRA Total Hours - everything logged in JIRA for that period

Leave / Holiday Hours - hours in JIRA logged against any task whose name contains the word 'leave' or 'holiday' (e.g. Bank Holiday, Annual Leave)

Four rules are applied in order:

APPROVE Rule 1 - Hours match, no leave or holiday in JIRA

The employee submitted 40 hours. JIRA shows 40 hours. None of those hours are against a leave or holiday task. Everything checks out. Decision: APPROVED.

REJECT Rule 2 - Hours match, but leave or holiday IS in JIRA

The employee submitted 40 hours. JIRA shows 40 hours - but 8 of those hours are logged against "Bank Holiday". Leave time should not be submitted in Sparta. Decision: REJECTED.

APPROVE Rule 3 - Hours differ, but the gap is exactly explained by leave

The employee submitted 32 hours. JIRA shows 40 hours - 8 of which are on a Bank Holiday task. The gap between JIRA and Sparta is exactly 8 hours. The employee correctly excluded their leave from Sparta. Decision: APPROVED.

REJECT Rule 4 - Hours differ and leave does not explain the gap

The employee submitted 35 hours. JIRA shows 40 hours - 8 are on Annual Leave. The gap is 5 hours, but leave accounts for 8 hours. The numbers are inconsistent. Decision: REJECTED.

Decision Summary Table

Sparta vs JIRA	Leave/Holiday in JIRA?	Gap = Leave Hours?	Decision
Hours match	No	-	APPROVE
Hours match	Yes	-	REJECT
Hours differ	Yes	Yes	APPROVE
Hours differ	Yes	No	REJECT

5. The Audit Report

A PDF audit report is automatically generated for every single timecard processed. Reports are named after the timecard ID, for example:

audit_TCH-05-14-2026-335964.pdf

Each report is uploaded and a download link is provided so it can be stored, shared, or reviewed at any time. The report contains four sections:

Section 1 - Timecard Details

A summary table of the timecard: employee name, period, hours submitted to Sparta, project name, decision reached, and the date it was processed.

Section 2 - JIRA Work Log Table

Every JIRA log entry found for the employee and period, showing the date, JIRA issue key, task name, and hours logged. Leave and holiday entries are clearly highlighted.

Section 3 - Hour Calculation

A clear breakdown showing:

- Total hours logged in JIRA
- Hours logged against leave or holiday tasks
- Net project hours (JIRA total minus leave/holiday)
- Hours submitted in Sparta
- The gap between JIRA and Sparta
- Whether the gap is fully explained by leave

Section 4 - Decision

States the exact rule that was applied and the final decision, displayed prominently in green (APPROVED) or red (REJECTED). This is the definitive record of why a timecard was approved or rejected.

6. The Audit Log

In addition to individual PDF reports, a running Audit Log (AUDIT_LOG.md) is maintained in the GitHub repository (RDG-QA/RDG---QA---Sparta-Timesheet-Approver). This is updated automatically every time a timecard is processed and contains a full history of every decision made, including:

- Date processed
- Timecard ID
- Employee name
- Period covered
- Sparta hours vs JIRA hours
- Leave/holiday hours identified
- Decision and rule applied

This provides a complete, searchable record for compliance or review purposes.

7. Fortnightly Schedule

The tool is set to run on a fortnightly cycle:

- A WhatsApp reminder is sent to the manager every 14 days at 9am, prompting them to provide the resource name and initiate the timecard processing run.
- The GitHub repository (containing all scripts and this manual) is automatically synced every 14 days to ensure it stays up to date.
- The tool will not start without the manager first providing the resource name - ensuring every run is intentional and correctly scoped.

8. Contact

For questions about this process or the tool itself:

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End of User Manual - Sparta Timesheet Approver v1.0 - Rail Delivery Group - June 2026